

Sai Kiran Yerramsetty

Senior Full-Stack Engineer & AI/GenAI Integration Specialist

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PROFESSIONAL SUMMARY

Senior Full-Stack Engineer and AI/GenAI Integration Specialist with **6+ years** of production experience building scalable applications across fintech, insurance, and SaaS domains. Available for freelance and contract engagements - remote-friendly. Expert across the complete modern stack: **React, Angular, Next.js, Node.js, Java (Spring Boot), Python, TypeScript, and GraphQL**. Deep AI/GenAI capabilities including **OpenAI API integration, Retrieval-Augmented Generation (RAG), conversational AI interfaces, and AI-powered automation workflows**. Cloud-native practitioner on **AWS and Azure** with Kafka, Docker, Kubernetes, and Terraform experience at enterprise scale (2,500+ req/min, 1M+ events/day). Databases across SQL and NoSQL: **PostgreSQL, MySQL, MongoDB, Redis, Oracle, Firebase**. Testing and automation: **Cypress, Playwright, Selenium, Jest, JUnit, BDD/Cucumber**.

- Full-Stack Web Apps
- AI / GenAI Integration
- REST & GraphQL APIs
- Microservices Architecture
- Cloud Migration (AWS / Azure)
- Python Automation
- Performance Optimization
- CI/CD & DevOps
- RAG Pipelines
- Test Automation

CORE SKILLS

- React.js
- Angular (v14+)
- Next.js
- TypeScript
- JavaScript ES6+
- Redux Toolkit
- Tailwind CSS
- Material UI
- Figma
- WCAG Accessibility
- Micro-frontend
- Node.js
- Java
- Spring Boot
- Spring Cloud
- Python
- Express.js
- GraphQL
- REST APIs
- WebSockets
- Socket.IO
- OpenAI API
- GenAI Integration
- RAG Pipelines
- Conversational AI
- AI Automation
- LLM Workflows
- GitHub Copilot
- AWS Lambda
- AWS EC2 / S3 / RDS
- AWS DynamoDB
- Azure
- Docker
- Kubernetes
- Terraform
- CloudFormation
- GitHub Actions
- Jenkins
- PostgreSQL
- MySQL
- MongoDB
- Redis
- Oracle
- Firebase
- OpenSearch
- Apache Kafka
- RabbitMQ
- Event-Driven Architecture
- Cypress
- Playwright
- Selenium
- Jest
- JUnit
- BDD / Cucumber
- Mockito

TECHNICAL SKILLS

Frontend	React.js, Angular (v14+), Next.js, TypeScript, JavaScript (ES6+), Redux Toolkit, Context API, Material UI, Tailwind CSS, Webpack, Babel, HTML5, CSS3, Figma, Micro-frontend Architecture, WCAG Accessibility, Storybook, Responsive Design
Backend	Node.js, Java, Spring Boot, Spring Cloud (Eureka, Config Server), Python, Express.js, GraphQL, RESTful APIs, Microservices, Domain-Driven Design (DDD), WebSockets, Socket.IO, Event-Driven Services
AI & GenAI	OpenAI API Integration, Retrieval-Augmented Generation (RAG), LLM Workflow Design, Conversational AI Interfaces, AI-Powered Automation, Intelligent Data Classification, AI-Driven Log Analysis, Anomaly Detection, GitHub Copilot, AI-Assisted Code Generation
Cloud & DevOps	AWS (EC2, S3, Lambda, RDS, DynamoDB, CloudWatch), Azure, Docker, Kubernetes (Helm), Terraform, CloudFormation, GitHub Actions, Jenkins, Grafana, CI/CD Pipelines, Infrastructure as Code
Databases	PostgreSQL, MySQL, Oracle, MongoDB, Firebase Firestore, Redis, DynamoDB, OpenSearch, JPA / Hibernate, SQL, PL/SQL, Query Optimization, Spring Data JPA, Spring Batch
Messaging & Streaming	Apache Kafka (producers, consumers, partitions, consumer groups), RabbitMQ, Redis Pub/Sub, OpenSearch, Distributed Messaging, Event-Driven Architecture
Testing & Automation	Jest, Mocha, Cypress, Playwright, Selenium, JUnit, Mockito, BDD Methodology, Cucumber, Postman, Performance Testing (load, stress, latency), SonarQube, npm audit
Security	JWT, OAuth2, Spring Security, Keycloak SSO, Role-Based Access Control (RBAC), GDPR / PCI Compliance
Tools & Process	Git, Jira, Confluence, Agile / Scrum, Storybook, Gradle, Groovy (Jenkins pipelines), Figma

WORK EXPERIENCE

American Express

Aug 2024 - Present

Senior Full Stack Developer

Dallas, TX

- Architected and delivered scalable full-stack customer analytics and payment insights platform using **Angular, TypeScript, Node.js, Java Spring Boot**, and microservices supporting high-volume financial transaction data and real-time analytics dashboards.
- Designed 15+ reusable UI components and enterprise dashboards using Angular, TypeScript, Material UI; reduced frontend dev effort by 40% and achieved WCAG accessibility compliance.
- Optimized frontend performance using Webpack, Babel, lazy loading, and code splitting - load time 4.2s to 1.8s, Lighthouse score to 92.
- Developed backend services using **Java Spring Boot and Spring Cloud** (Eureka, Config Server) along with **Node.js** microservices with domain-driven design (DDD).
- Built and maintained 12+ high-performance **REST APIs** supporting 2,500+ requests/min with sub-200ms latency; implemented **GraphQL** endpoints for analytics workflows.
- Implemented data access layers with **JPA/Hibernate + PostgreSQL**; optimized queries and indexing for high-volume financial data.
- Led migration of a legacy monolithic **Python** service to distributed **Node.js** microservices - 3.5x throughput improvement, independent scaling enabled.
- Designed **Kafka**-based event-driven architecture; developed producers and consumers for transaction events, analytics pipelines, and notifications - 1M+ events/day with fault tolerance.
- Deployed containerized services using **Docker and Kubernetes** (Helm) across AWS and Azure; automated **CI/CD** with GitHub Actions reducing release time from 25 min to 3 min.
- Implemented infrastructure automation with **Terraform**; observability with Grafana and CloudWatch; centralized error handling reducing production exceptions by 80%.
- Built serverless document processing pipelines using **AWS Lambda + S3** processing 8,000+ documents daily; real-time notification systems supporting 1.2M+ daily events.
- Implemented secure coding: **JWT, OAuth2, RBAC** for enterprise security compliance.
- Achieved 93% test coverage using Jest, Mocha, and JUnit; identified 120+ critical pre-production issues.
- Introduced **Storybook**-based design system, reducing UX review cycles by 50%.

AI / GENAI ENGINEERING

- Integrated **OpenAI APIs** to automate documentation generation, customer insight summarization, and intelligent data classification across analytics platforms.
- Designed **Retrieval-Augmented Generation (RAG)** architecture combining indexed transactional datasets with REST APIs for contextual AI-powered search and recommendations.
- Built **conversational analytics interfaces** in Angular enabling natural language queries and contextual insights for internal business users.
- Implemented **AI-driven log analysis and anomaly detection** pipelines to accelerate root-cause identification during production incidents.
- Leveraged **GitHub Copilot** and AI-assisted workflows to accelerate code generation, refactoring, and test case creation.

Manomay Insurtech

Jul 2021 - Aug 2023

Full Stack Developer

Hyderabad, India

- Built insurance SaaS platform end-to-end: claims management, policy dashboards, agent onboarding, and real-time customer communication tools.
- Built claims management system using **React (Hooks, Redux Toolkit)** with Formik + Yup forms - input errors reduced 40%; real-time notifications via **WebSockets / Socket.IO** - user engagement up 35%.
- Built live chat module using **Socket.IO**, reducing customer support resolution time by 25%.
- Developed interactive dashboards using **Chart.js and D3.js** for real-time claims and policy data visualization.
- Configured **Redis** caching (60% API response time reduction), Redis Pub/Sub for real-time broadcasts, rate limiting for API abuse prevention, and Redis session store for 10K+ concurrent auth sessions.
- Built REST APIs in **Node.js / Express** for claims validation, onboarding, and reporting; secured with **Spring Security, JWT, and OAuth2** (GDPR/PCI compliant).
- Migrated monolithic **Java** application to microservices - modularity improved, release time reduced 35%.
- Integrated **MongoDB** for document storage and **MySQL** for transactional workflows; tuned SQL queries, stored procedures, and triggers - DB response time reduced 45%.

Capita India Pvt Ltd

Apr 2020 - Jun 2021

Software Associate - Full Stack Developer

Bengaluru, India

- Built **React** HR dashboards with Material-UI, charts, and grids - HR reporting time cut from 3 days to 1 hour for 1,000+ users.
- Implemented **Redux Toolkit** state management and **React.lazy()** dynamic imports improving page load by 45%; reusable Formik forms reduced code duplication by 35%.
- Built **Spring Boot** microservices for HR, payroll, and attendance automation; integrated Spring Data JPA + Hibernate with **Oracle DB**; Spring Batch jobs reduced data migration time by 40%.
- Secured services with **Spring Security + Keycloak SSO**; implemented **Redis** caching (45% latency reduction).
- Orchestrated deployments with **Docker + Kubernetes on AWS EKS** (downtime under 2 min); Jenkins CI/CD improved release frequency from monthly to weekly.
- Configured **JUnit, Jest, Mocha, Cypress** for testing (90%+ coverage); automated regression suites with **Cypress + GitHub Actions** - manual QA time down 50%.

CGI

Feb 2018 - Mar 2020

Software Engineer - Full Stack

Bengaluru, India

- Developed responsive **React, Redux, TypeScript** UI components and real-time financial dashboards using **WebSockets + Chart.js**.
- Migrated legacy AngularJS/jQuery apps to **Angular and React** - technical debt reduced 40%.
- Published custom **npm packages** for reusable React form controls - duplicate code across teams reduced 25%.
- Built REST APIs with **Java Spring Boot**; integrated **Oracle + PostgreSQL** for reporting; designed SQL triggers and stored procedures (batch processing 30% faster).
- Designed **Jenkins** declarative pipelines (Groovy) cutting release cycle 40%; automated **Terraform** infrastructure provisioning; integrated SonarQube reducing vulnerabilities by 30%.

PERSONAL PROJECTS

POS System for Retail Billing (PoC) [React](#), [Node.js](#), [Firebase](#), [Figma](#), [Material-UI](#)

- Designed 12+ billing interfaces in **Figma** and built them with **React + Material-UI** - staff training time reduced 30%.
- Built prototype POS dashboard with real-time transaction tracking - billing time cut from 90s to 35s per customer.
- Engineered **Node.js RESTful APIs** for billing operations: 50+ transactions/min, <1% error rate under stress testing.
- Leveraged **Firebase Firestore** for real-time transaction sync (200ms read/write latency across 10+ demo stores); composite indexing reduced monthly Firebase costs by 40%.

EDUCATION

Master of Science, Management and Information Technology

2023 - 2025 | GPA: 3.87 / 4.0

[Indiana Wesleyan University](#)

Bachelor of Commerce (B.Com)

2014 - 2017 | GPA: 3.9 / 4.0

[Gandhi Institute of Technology and Management](#)

CERTIFICATIONS

AWS Certified Developer - Associate [Amazon Web Services](#)

Azure Developer Associate [Microsoft](#)

ACHIEVEMENTS

- Professional para table tennis athlete - competed in 3 Para National Championships (2021, 2022, 2023), ranked top 16 nationally.
- Texas State Table Tennis Championship participant.
- State-level table tennis competitor, India.